

Name _____

Period _____ Date _____

Practice Test – Advanced Kinematics (Version 2)

On any problems, you must show all work, include units, round properly, and box your final answer.

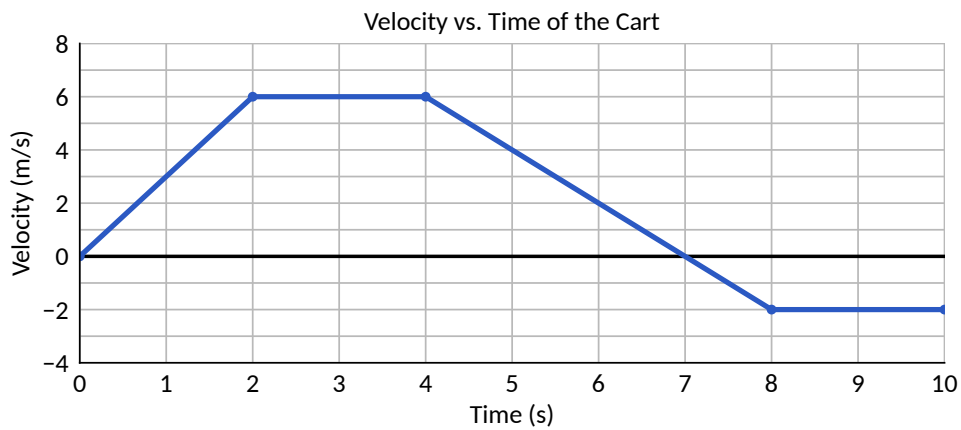
Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

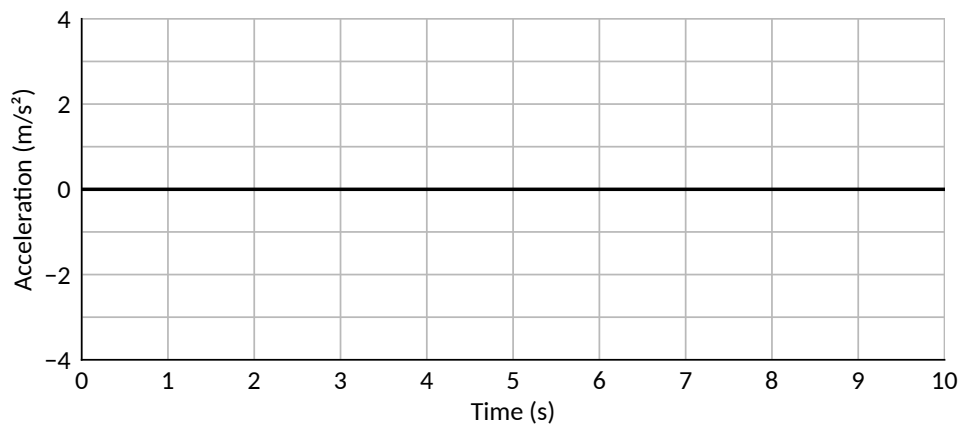
For all questions:

- Use $g = 9.80 \text{ m/s}^2$ and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).
- The last page is optional extra practice and is not part of the timed test.

1. (25 points) The velocity-time graph below shows the motion of a cart moving along a straight track from $t = 0$ s to $t = 10$ s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

2. (25 points) A car travels along a straight, level road at a constant 27.0 m/s. The driver then brakes, and the car slows with a constant acceleration of magnitude 4.50 m/s^2 until it stops. How far does the car travel while braking?

3. (25 points) A football is kicked from ground level on a level field with an initial speed of 22.0 m/s at 40.0° above the horizontal. It lands at the same height from which it was kicked. Air resistance is negligible. What is the ball's range (the horizontal distance it travels before landing)?

4. (25 points) A small object of mass m is released from rest at the roof of a skyscraper, a height H above the street, as shown in Figure 1. The object falls straight down to the street. Air resistance is negligible.

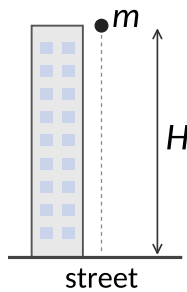


Figure 1

A. Derive an expression for v , the speed of the object just before it hits the street. Express your answer in terms of m , H , and physical constants, as appropriate. Begin your derivation by writing a fundamental physics principle or an equation from the reference information.

B. A second object of mass $2m$ is released from rest at the same roof. Let v_2 be the speed of the second object just before it hits the street. Indicate whether v_2 is greater than, less than, or equal to v by writing one of the following.

- $v_2 > v$ • $v_2 < v$ • $v_2 = v$

Briefly justify your answer by referencing your derivation in part A.

Optional extra practice (not part of the timed test, not scored)

E1. For the football in Question 3: (a) What is its hang time? (b) What is its maximum height?

E2. For the car in Question 2: (a) How long does it take to stop? (b) If its initial speed were doubled and it braked with the same acceleration, by what factor would its braking distance change?

E3. For the object in Question 4: write its velocity $v_y(t)$ as a function of the time t since it was released, and state the time interval over which this expression is valid.

Answer Key – Practice Test, Version 2 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units. $g = 9.80 \text{ m/s}^2$; answers to three significant figures.

AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):

#1 $v_x = v_{x0} + a_x t$ · #2 $x = x_0 + v_{x0}t + \frac{1}{2}a_x t^2$ · #3 $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Same equations as $v_f = v_i + at$, $\Delta x = v_i t + \frac{1}{2}at^2$, $v_f^2 = v_i^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need. For vertical motion use y in place of x .

1. Acceleration-time graph (25 points)

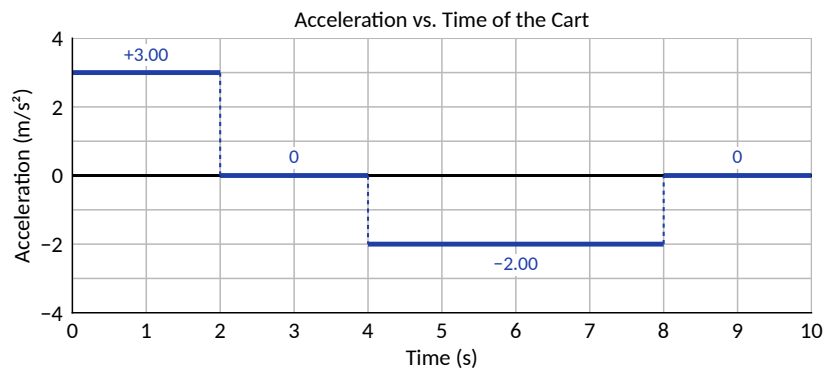
Step 1 Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece; values read off grid lines are treated as exact)}$$

Step 2 Find the slope of each segment.

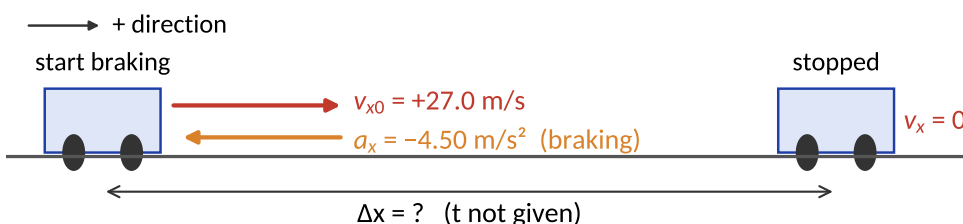
segment	$\Delta v = v_f - v_i$	$\Delta t = t_f - t_i$	$a = \Delta v / \Delta t$
0 s to 2 s	6 m/s - 0 m/s = 6 m/s	2 s - 0 s = 2 s	+3.00 m/s ²
2 s to 4 s	6 m/s - 6 m/s = 0 m/s	4 s - 2 s = 2 s	0 m/s ²
4 s to 8 s	(-2 m/s) - 6 m/s = -8 m/s	8 s - 4 s = 4 s	-2.00 m/s ²
8 s to 10 s	(-2 m/s) - (-2 m/s) = 0 m/s	10 s - 8 s = 2 s	0 m/s ²

Step 3 Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: the cart turns around where v crosses 0, but the slope does not change there, so a does not change either. A straight slanted v-t line always becomes a flat a-t line; a flat v-t line becomes $a = 0$.

2. Braking car: stopping distance (25 points)



Step 1 Pick a positive direction and get the sign of a.

Let the direction the car moves be +. The car is moving + but slowing down, so its velocity is decreasing: Δv is negative, which makes a negative. Braking acceleration points opposite the motion, so $a = -4.50 \text{ m/s}^2$.

Step 2 List the five variables.

$v_i (= v_{x0})$	$v_f (= v_x)$	a	t	Δx
+27.0 m/s	0 m/s (stops)	-4.50 m/s ²	not given, not asked	? (wanted)

Step 3 Pick the equation.

AP sheet	equation	leaves out
#1	$v_x = v_{x0} + a_x t$	$x - x_0$ (Δx)
#2	$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	v_x (final velocity)
#3	$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	t (time) ← use this one

Why #3: time is not given and not asked for. #3, written $v_f^2 = v_i^2 + 2a\Delta x$, is the only one without t.

Step 4 Start from $v_f^2 = v_i^2 + 2a\Delta x$ and set $v_f = 0$. Solve for Δx before any numbers.

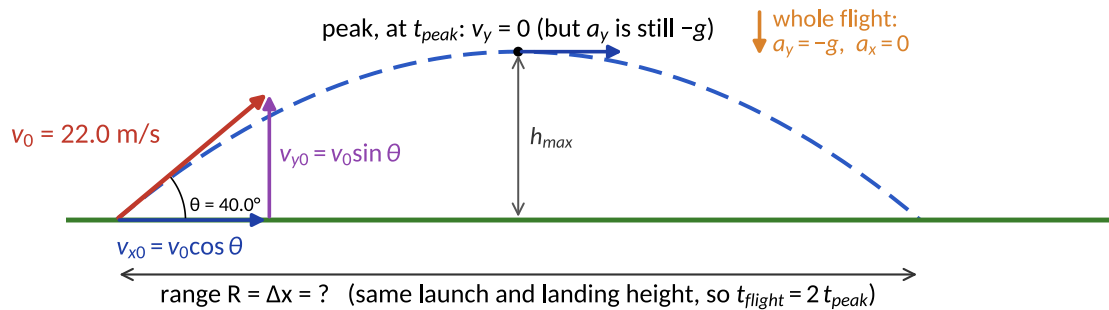
$$v_f^2 = v_i^2 + 2a\Delta x \rightarrow 0 = v_i^2 + 2a\Delta x \rightarrow 2a\Delta x = -v_i^2 \rightarrow \Delta x = -\frac{v_i^2}{2a}$$

Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+27.0 \text{ m/s})^2}{2(-4.50 \text{ m/s}^2)} = -\frac{(729 \text{ m}^2/\text{s}^2)}{(-9.00 \text{ m/s}^2)} = \boxed{81.0 \text{ m}}$$

The minus sign in the formula and the negative a cancel, so Δx comes out positive: the car moves forward while it stops. If you get a negative distance, a had the wrong sign. Units: $(\text{m}^2/\text{s}^2) \div (\text{m/s}^2) = \text{m}$.

3. Football: range (25 points)



Split the kick into two motions. Sideways moves at a constant speed ($a_x = 0$). Up and down is free fall ($a_y = -g$). **Time** is the only thing both share.

Step 1 Resolve the initial velocity into components (keep letters; numbers go in at the end).

$v_{x0} = v_0 \cos \theta$ (the side next to the angle: along the ground)

$v_{y0} = v_0 \sin \theta$ (the side across from the angle: straight up)

Step 2 Make the T-chart.

horizontal	vertical
$a_x = 0$	$a_y = -g = -9.80 \text{ m/s}^2$
$v_{x0} = v_0 \cos \theta$	$v_{y0} = v_0 \sin \theta$
$\Delta x = R = ?$	at the peak: $v_y = 0$
$t = t_{\text{flight}}$	$t_{\text{peak}} = ?$

Step 3 Find the time to the peak, where $v_y = 0$.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0 (\Delta y) \leftarrow \text{use this one}$
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2 a_y (y - y_0)$	t (time)

Why #1: at the peak you know $v_y = 0$, v_{y0} and a_y , and you want t . The height of the peak is not known, so use #1, the one that leaves out Δy .

$$v_y = v_{y0} + a_y t \rightarrow 0 = v_0 \sin \theta - g t_{\text{peak}} \rightarrow t_{\text{peak}} = \frac{v_0 \sin \theta}{g}$$

Step 4 Double it for the whole flight.

The ball lands at the same height it was kicked from, so the trip down takes exactly as long as the trip

$$\text{up: } t_{\text{flight}} = 2t_{\text{peak}} = \frac{2v_0 \sin \theta}{g}$$

Watch out: at the peak $v_y = 0$, but the acceleration is still $a_y = -g$, pointing down. Gravity never switches off. Doubling t_{peak} only works because the launch and landing heights match. And $\Delta y = 0$ over the whole flight means it ends at the height it started, not that the acceleration is zero.

Step 5 Use the flight time in the horizontal column.

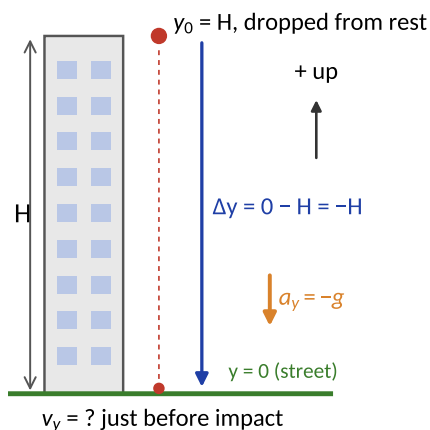
Why #2 (x-form): horizontally you know t and v_{x0} and want Δx ; with $a_x = 0$, #2 becomes $\Delta x = v_{x0}t$.

$$x - x_0 = v_{x0}t + \frac{1}{2}a_x t^2 \rightarrow R = v_{x0}t_{\text{flight}} = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} \rightarrow \boxed{R = \frac{2v_0^2 \sin \theta \cos \theta}{g}}$$

Step 6 Now plug in numbers with units.

$$R = \frac{2(22.0 \text{ m/s})^2(\sin 40.0^\circ)(\cos 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{48.6 \text{ m}}$$

Check: $t_{\text{peak}} = (22.0 \text{ m/s})(\sin 40.0^\circ)/(9.80 \text{ m/s}^2) = 1.443 \text{ s}$, $t_{\text{flight}} = 2.886 \text{ s}$, $v_{x0} = 16.85 \text{ m/s}$, and $(16.85 \text{ m/s})(2.886 \text{ s}) = 48.63 \text{ m} \approx 48.6 \text{ m}$. ✓ (2 sin θ cos θ = sin 2θ , so this also equals $v_0^2 \sin 2\theta / g$.)

4. Dropped from a skyscraper, AP style (25 points): A. derive (18), B. justify with it (7)

Reading the AP wording. "Physical constants" means g : you're expected to use it even though it's not listed. "As appropriate" means you may not need every letter listed. Here m drops out, because every object in free fall has the same acceleration g , whatever its mass.

A speed has no sign, so any convention works; up is + (class default). Origin at the street: $y_0 = H$, $y = 0$, so $y - y_0 = -H$. Gravity points down: $a_y = -g$.

A. Derivation (18 points)**Step 1 List the variables (vertical, so y in place of x).**

v_{y0} (initial)	v_y (final)	a_y	t	$y - y_0$
0 (released from rest)	? (wanted)	$-g$	not given, not asked	$0 - H = -H$

Step 2 Write the equation from the reference information first.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time) ← use this one

Why #3: t is not given and not asked. On the AP exam, write this equation on its own line before substituting anything; that line is where the first point is.

Start: $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$

Step 3 Substitute and solve with letters.

$$v_y^2 = 0^2 + 2(-g)(0 - H) = 2gH \rightarrow v_y = \pm\sqrt{2gH}$$

Step 4 Take the size, since speed is asked.

Speed is the magnitude of the velocity, so it is the positive value: $v = \sqrt{2gH}$ No m in it. Units:

$$\sqrt{(m/s^2)(m)} = m/s.$$

If the question had asked for **velocity**, you'd also give the direction: with up as +, $v_y = -\sqrt{2gH}$. Read whether it says speed or velocity.

B. Justify with your derivation (7 points)

$v_2 = v$ From part A, $v = \sqrt{2gH}$, which contains only g and H. Mass does not appear, so changing m to 2m changes nothing: both objects fall with acceleration g from the same height H and hit with the same speed.

Optional extra practice (not scored)

E1. Football: hang time and maximum height

$$(a) \text{ Hang time} = t_{\text{flight}} = \frac{2v_0 \sin \theta}{g} = \frac{2(22.0 \text{ m/s})(\sin 40.0^\circ)}{(9.80 \text{ m/s}^2)} = \boxed{2.89 \text{ s}}$$

Why #3 (y-form) for (b): at the peak $v_y = 0$, and you want the height without needing t .

$$(b) 0 = (v_0 \sin \theta)^2 - 2g h_{\text{max}} \rightarrow h_{\text{max}} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{(22.0 \text{ m/s})^2 (\sin 40.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{10.2 \text{ m}}$$

E2. Car: stopping time and the doubling question

Why #1: now t is wanted and Δx is not needed. $v_f = v_i + at \rightarrow 0 = v_i + at \rightarrow t = -\frac{v_i}{a} = -\frac{(+27.0 \text{ m/s})}{(-4.50 \text{ m/s}^2)} =$

$$\boxed{6.00 \text{ s}}$$

(b) $\Delta x = -\frac{v_i^2}{2a}$: v_i is squared, so doubling it multiplies Δx by $2^2 = \boxed{4}$ ($4 \times 81.0 \text{ m} = 324 \text{ m}$). Not 2.

E3. Skyscraper: velocity as a function of time

Why #1 (y-form): it links v_y and t directly. Up is +. $v_y = 0 + (-g)t \rightarrow \boxed{v_y(t) = -gt}$

Why #2 (y-form) for the impact time: you know $y - y_0 = 0 - H = -H$ and want t . $-H = -\frac{1}{2}gt_f^2 \rightarrow t_f =$

$\sqrt{2H/g}$, so it is valid for $\boxed{0 \leq t \leq \sqrt{2H/g}}$. Check: $-g\sqrt{2H/g} = -\sqrt{2gH}$, matching Question 4.

Suggested partial credit (25 points per question, 100 total)

1. Acceleration-time graph

Suggested partial credit	pts
Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each)	8
Correct slope value with sign and units for each segment (1 each)	4
Four horizontal segments at the correct accelerations (2 each)	8
Changes in acceleration occur at the correct times; no slanted lines	2
Axes labeled with quantity and units; readable scale	3
Total	25

2. Braking car

Suggested partial credit	pts
Starts from $v_f^2 = v_i^2 + 2a\Delta x$ (sheet #3) because time is not involved	5
Sets $v_f = 0$	3
Acceleration negative, with a reason (opposite the motion / slowing down)	4
Solves $\Delta x = -v_i^2 / (2a)$ before substituting	6
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

3. Football range

Suggested partial credit	pts
Resolves v_0 into $v_0 \cos \theta$ and $v_0 \sin \theta$	4
Time to peak from $v_y = 0$ with sheet #1: $t_{\text{peak}} = v_0 \sin \theta / g$	5
Doubles t_{peak} for the flight time, with the equal-height reason	4
$R = v_{x0}t$ with the expression for t substituted (not a rounded number)	5
Substitution with units in every parenthesis	4
Final answer, 3 significant figures, units, boxed	3
Total	25

4. Skyscraper impact velocity

Suggested partial credit	pts
Writes an equation from the reference information first: $v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	4
$v_{y0} = 0$; a_y and $y - y_0$ with signs consistent with a stated convention	5
Algebra to $v_y^2 = 2gH$	4
Gives the speed as the positive magnitude, $v = \sqrt{2gH}$, boxed	3
Answer uses only H and g (m correctly left out)	2
B: indicates $v_2 = v$	2
B: justifies from the part A expression (m does not appear in $v = \sqrt{2gH}$)	5
Total	25